

Lecture 10:

AI504 - Knowledge Representation

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1 Propositional logic

Remember...

- a signature is just a set (“sentence letters”)
- if P is a signature, a P -sentence is generated by recursion as follows

$$(p \vee \neg q) \Leftrightarrow (r \wedge p)$$

These can be broken into

$$p, q, r$$

Base case: a single

- letter $\in P$ is a sentence.
- Boolean constants $(T, F)/(\top, \perp)$

Additionally

- if ϕ is a sentence then so is $\neg\phi$
- if ϕ and ψ are sentences, so are

$$\phi \wedge \psi, \phi \vee \psi, \phi \Rightarrow \psi, \phi \Leftrightarrow \psi$$

A P -model

A P -model is a function. $P \rightarrow \{\top, \perp\}$ (sometimes called a truth assignment/valuation)

$$\mathcal{M} = (P, P \rightarrow \{\top, \perp\})$$

if $\mathcal{M}$ is a model, and ϕ is a sentence in the same signature, define in the same signature (P), $\llbracket \phi \rrbracket \in \{\top, \perp\}$ as follows.

If ϕ is a single letter *e.g.* p , then $\llbracket \phi \rrbracket$ is defined by recursion on construction of ϕ (where the base case $p \in P$ is given by $\mathcal{M}$)

1.1 example

for $P = \{p, q, r\}$ and $\mathcal{M} = \{(p, \top), (q, \top), (r, \perp)\}$ where for $\phi = (r \wedge \neg p) \vee q$

then

$$\llbracket \phi \rrbracket = \llbracket ((r \wedge \neg p) \vee q) \rrbracket = \llbracket r \wedge \neg p \rrbracket \vee \llbracket q \rrbracket = \top$$

2 Tautologies

For “Does $\emptyset \models (p \vee q) \wedge (p \vee r) \Rightarrow p \vee (q \wedge r)$ ”

A general fact applies

Let $\bigwedge \Gamma$ be the conjunction of all sentences in Γ

Then $\Gamma \models \phi$ is the same as $\bigwedge \Gamma \Rightarrow \phi$ being a tautology

3 Contradictions

Similarly $\{\psi\} \models F$

This means that every models satisfying ψ must satisfy F (contradiction)

4 Logical equivalence

ϕ and ψ are logically equivalent if for all models $\mathcal{M}$,

$$\phi \equiv \psi$$

$$\mathcal{M} \models \phi \iff \mathcal{M} \models \psi$$

4.1 equivalent formulations

- $\phi \iff \psi$ is a tautology
- $\{\phi\} \models \psi$ and $\{\psi\} \models \phi$

5 Good to Remember

$$p \vee \neg p \equiv \top$$

$$p \wedge \neg p \equiv \perp$$

$$\neg \neg \phi \equiv \phi$$

$$\neg(\psi \wedge \phi) \equiv (\neg\psi) \vee (\neg\phi)$$

$$\neg(\psi \vee \phi) \equiv (\neg\psi) \wedge (\neg\phi)$$